

Fixed-outline constraints

Floorplanning on this course uses a fixed outline ten by eight

Pseudocode

- Fixed-outline legality is two loops in pseudocode
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Outline is ten by eight
- Golden packing of A through E is legal
- Bad seed with E at x nine overflows
- Overlap of C and E is the other failure mode

Algorithm sketch — try these

```
INPUT: outline  $W \times H$ , modules  $(x, y, w, h)$   
OUTPUT: legal? / failure reason  
for each m: fail if outside  $[0, W] \times [0, H]$   
for each pair: fail if interior overlap  
edge-touch OK; positive-area overlap not  
GOLDEN pack legal; E@x=9 overflow illegal  
outline  $10 \times 8$  (area 80); modules A–E
```

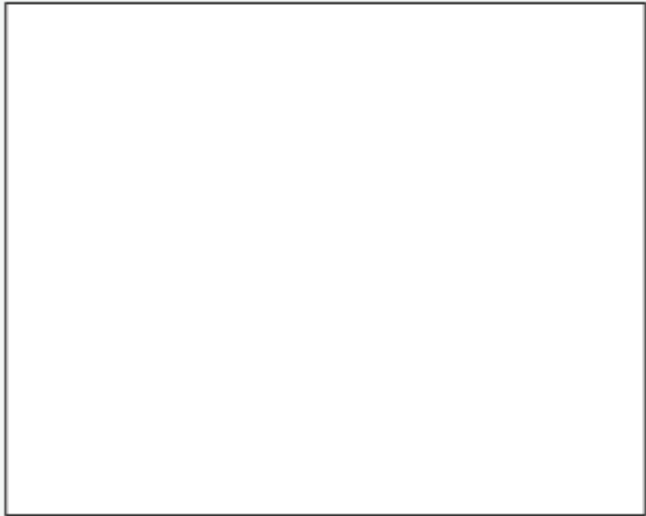
Fixed outline is 10×8

FIXED OUTLINE

Fixed outline is 10×8

Step 1 / 5 · outline · module01-01-fixed-outline

10×8



Explanation

Modern floorplanning starts with a fixed outline. Ours is ten by eight—area eighty. Modules A through E must pack inside; growing the chip is not allowed.

- Outline $W \times H = 10 \times 8$
- Lower-left coordinates
- Legality before wirelength

```
outline: 10×8  
outlineArea: 80  
modules: A–E
```

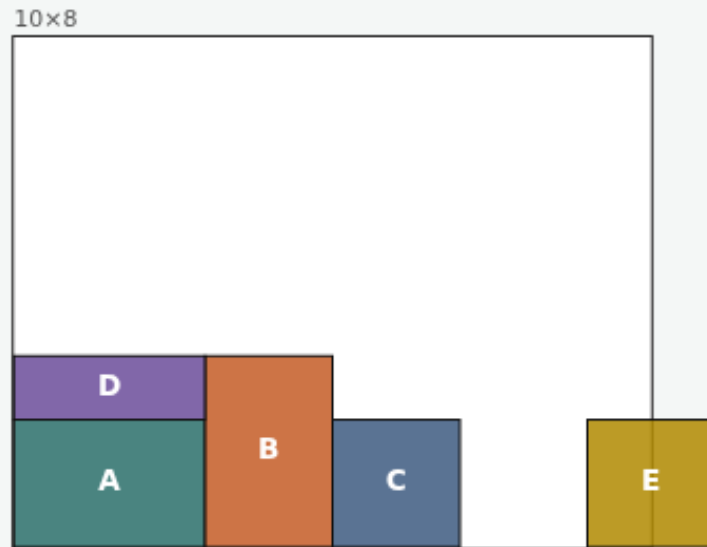
Fixed outline is 10×8

Bad pack: E overflows

FIXED OUTLINE

Bad pack: E overflows

Step 2 / 5 · bad-overflow · module01-01-fixed-outline



Explanation

The bad seed places E at x equals nine with width two, so it sticks past the right edge. Legality fails immediately—E is outside the outline.

- E: $x=9, w=2 \rightarrow$ right edge 11
- Outline width is only 10
- Illegal before density matters

legal: false
reason: E outside outline
deadspace (areas): 57

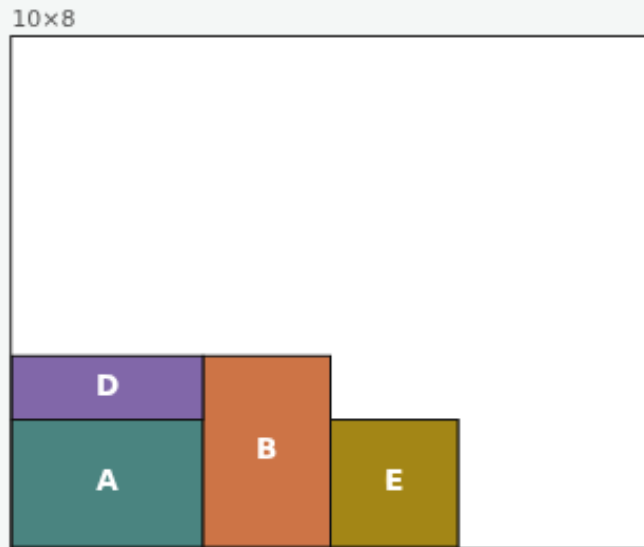
Bad pack: E overflows

Overlap is also illegal

FIXED OUTLINE

Overlap is also illegal

Step 3 / 5 · overlap · module01-01-fixed-outline



Explanation

A second failure mode: E sits on top of C. Edge-touching is fine; positive-area interior overlap is not. The checker reports C overlaps E.

- Interior overlap rejected
- Edge touching allowed
- Same outline, wrong packing

legal: false
reason: C overlaps E

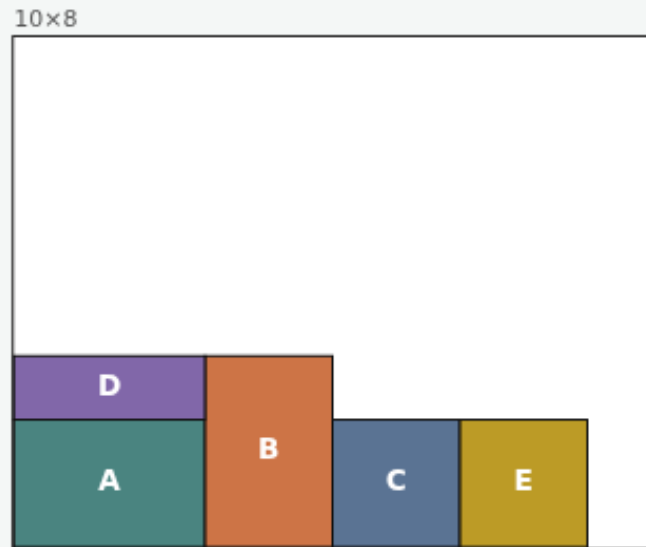
Overlap is also illegal

Golden pack is legal

FIXED OUTLINE

Golden pack is legal

Step 4 / 5 · golden · module01-01-fixed-outline



Explanation

The golden packing keeps every block inside ten by eight with no overlaps. A sits at the origin; B, C, and E march right; D stacks above A.

- A at (0,0), D above A
- B-C-E along the bottom
- legal: true, reason: ok

legal: true
reason: ok
report: ok

Golden pack is legal

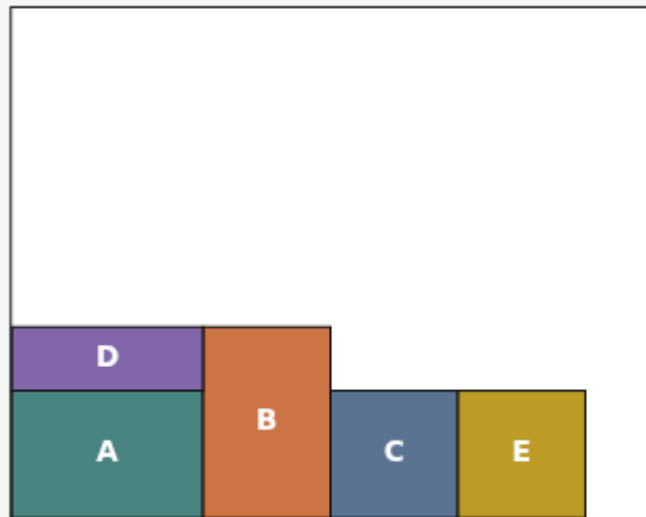
Legality is the gate

FIXED OUTLINE

Legality is the gate

Step 5 / 5 · takeaway · module01-01-fixed-outline

10×8



Explanation

Never optimize density or wirelength on an illegal packing. Fix containment and overlap first—then the metrics in the next lab mean something.

- Containment + non-overlap
- Golden: legal inside 10×8
- Next: deadspace 57 on this instance

```
golden legal: true  
bad legal: false
```

Legality is the gate

Browser lab track

- Open the fixed-outline lab
- Show the bad seed: E overflows past the right edge
- Then show overlap, then golden
- Watch legality flip from false to true when the golden packing loads

Implement track

- Parse `tiny_modules.json`, assign (x, y)
- On the golden packing, legality must pass

Pitfalls

- Off-by-one edges, treating centers as rectangles
- Edge-touching is allowed; interior overlap is not

Your turn

- Get a legal packing of A through E inside ten by eight
- Quiz checks the overflow failure and the golden pass
- Next: deadspace fifty-seven on this same instance